SCHOOL OF COMPUTING TECHNOLOGY AND APPLIED SCIENCES

# CCS4711 CRYPTOGRAPHY AND APPLICATIONS

**SEMESTER FINAL EXAMINATION**

**WEDNESDAY, 27 MAY 2026**

**09:00 – 12:00 HOURS**

TIME ALLOWED : WRITING – THREE HOURS : READING – 5 MINUTES

## INSTRUCTIONS:

1. Section A: this question is compulsory and must be attempted.
2. Sections B: Answer THREE (3) questions from this section.
3. This examination paper carries a total of 100 marks.
4. Candidates must not turn this page until the invigilator tells them to do so

**Section A**

**Question 1**

a) A university online examination system and a remote invigilation server need to establish a shared secret key before exchanging encrypted examination data. To achieve this, the two systems use the Diffie–Hellman key exchange algorithm.

The two systems agree on the following public values:

- P=17
- G=3

The university online examination system selects 5 as its private key, while the remote invigilation server selects 7 as its private key.

As a systems security analyst, compute:

a) The public key generated by the university online examination system.
b) The public key generated by the remote invigilation server.
c) The shared secret key established between the two systems.

Show all your workings. **(10 Marks)**

b) A pharmacy inventory system uses the RSA algorithm to protect short messages sent between a pharmacy branch and the central stock management server. For one communication session, the system is configured with the following values:

- p=13
- q=7
- e= 21
- d= 5

The system needs to send the three ASCII characters:

- $ = 36
- % = 37
- & = 38

Using the RSA algorithm, answer the following:

a) Calculate the modulus n.
b) Determine the public key.
c) Determine the private key.
d) Encrypt the ASCII values 36, 37, and 38 using the public key.
e) Decrypt the resulting ciphertext values using the private key.
f) Show that the original ASCII values are recovered after decryption.

Show all your workings. **(15 Marks)**

c) A land registry office wants to issue electronic land title certificates so that banks, buyers, and legal practitioners can verify that the certificates are authentic before property transactions are completed.

The office is considering a secure digital solution that may involve digital signatures, blockchain technology, and appropriate blockchain tools or platforms to support trust, verification, ownership tracking, and protection against tampering.

As a systems security analyst, design a secure electronic land title certificate verification system and explain how the land registry office, landowners, banks, buyers, and legal practitioners would interact with the system. **(8 marks)**

d) Explain the concepts of confusion and diffusion in cryptography and discuss how each principle helps to strengthen a cryptographic algorithm and give one example of how confusion and diffusion may be achieved in a block cipher. **(7 marks)**

**[Total 40 Marks]**

**Section B**

**Question 2**

a) A wildlife conservation agency wants to send short confidential messages between its field officers and headquarters. To protect the messages from casual interception, the agency decides to use the Playfair cipher, an early symmetric encryption technique that encrypts text by substituting pairs of letters called digraphs.

For one communication session, the agency uses the keyword:

FOREST

as the cipher key.

The field officer needs to encrypt the following message:

"Guard the wildlife reserve"

Using the Playfair cipher method, encrypt the message. **(10 marks)**

b) A small online electronics shop is considering accepting Bitcoin as a payment option from customers. The shop owner has heard that cryptocurrency transactions do not require a bank, can be verified on a blockchain, and may change in value quickly.

As an information systems student, explain how cryptocurrency would work in this situation and discuss what the shop owner should understand before accepting Bitcoin payments.

**(10 marks)**

**[Total 20 Marks]**

**Question 3**

a) A health insurance company wants to protect patients' medical claim records before storing them in a secure database. Each record contains the patient ID, diagnosis code, treatment details, and payment information. The company decides to use a block cipher design based on a Substitution-Permutation Network (SPN) because it provides security by repeatedly applying substitution and permutation operations over several rounds.

As a systems security analyst, explain how a Substitution-Permutation Network can be applied to encrypt the patients' medical claim records.

**(7 marks)**

b) A delivery company uses a mobile app for drivers to send parcel status, customer confirmation codes, and GPS location updates to a central server.The company wants to secure this data during transmission using both symmetric and asymmetric encryption. As a systems security analyst, explain how symmetric and asymmetric encryption can be practically used to secure communication between the driver's mobile app and the central server. **(7 marks)**

c) Modern cryptographic systems such as RSA require very large prime numbers to generate secure keys. Before a number is used as a prime in a cryptographic algorithm, it must be tested to determine whether it is likely to be prime or composite. Explain and demonstrate how a primality test can be used to check whether a number is suitable for use in RSA key generation. **(6 marks)**

**[Total 20 Marks]**

**Question 4**

a) A vehicle tracking company wants to improve the security of location updates sent from tracking devices to a central monitoring server. Each message contains sensitive data such as the vehicle ID, timestamp, latitude, longitude, and speed. The company decides to use a Feistel network as part of its encryption process. Explain and demonstrate how a Feistel network works in cryptography. **(7 marks)**

b) A fintech company wants to build a blockchain-based payment platform that allows customers to send and receive digital payments securely. The company is comparing two well-known blockchain consensus algorithms: Proof of Work and Proof of Stake.

As a blockchain consultant, explain how each consensus algorithm would work in the proposed payment platform and advise the company on which one would be more suitable. **(7 marks)**

c) RSA decryption can be slow when large numbers are used. To improve performance, RSA can use the Chinese Remainder Theorem (CRT) to speed up the decryption process. Explain how the Chinese Remainder Theorem is used to make RSA decryption faster. **(6 marks)**

**[Total 20 Marks]**

**Question 5**

a) A school examination office wants to send simple coded messages to invigilators during examination periods. To keep the messages hidden from students, the office decides to use a shift cipher, where each letter in the message is shifted by a fixed number of positions in the alphabet.

For one message, the examination office uses a shift key of:

K=4

The message to be encrypted is:

"SEND EXAM PAPERS"

As a cybersecurity student, demonstrate the process of using a shift cipher to encrypt and decrypt the message.

Your answer should show how each letter is shifted, produce the ciphertext, and then demonstrate how the original plaintext can be recovered. **(10 marks)**

b) A health insurance company has developed an online portal where clients can submit claims, upload medical documents, check claim status, and update their personal information. The portal is used by clients, claims officers, system administrators, and external health service providers.

Recently, the company discovered that some users were sharing passwords, some documents were accessed by unauthorised staff, and there was no proper record of who modified claim information.

As a cybersecurity student, explain how the company can apply the main principles of confidentiality, integrity, availability, authentication, authorisation, accountability, and non-repudiation and show how each principle can be applied practically in this system.

**(10 marks)**

**[Total 20 Marks]**

**END OF EXAMINATION**